Week 04 Reading: Extreme value theory

The GEV, the GPD, and why block maxima and exceedances give different answers

2026-09-14

Submit your questions on Canvas by Sunday 9/13 at 11:59 pm, before Monday's class.

Assigned

1. The HEC "Extreme Value Theory" video series from the USACE course *Statistical Methods in Hydrology*. Watch the YouTube playlist (four parts plus a key-takeaways video, about 90 minutes total). The downloadable slides with speaker notes read standalone if you prefer to read rather than watch. Focus on Parts I–III (slides 1–33): order statistics, the Fisher-Tippett-Gnedenko theorem, the GEV distribution, and convergence. Part IV (slides 34–48) introduces the GPD and peaks over threshold, which we cover fully in week 5; watch it now for context or save it for next week.
2. Section 3.1 of Ghil et al. (2011), which states the extremal types theorem and the three GEV families in the context you already know from weeks 2 and 3. This reading is open access.

Notation warning. The HEC deck uses κ for the GEV and GPD shape parameter. Most statistics texts, including Coles (2001) and Ghil et al. (2011), use ξ. They are the same parameter; the sign convention also differs in some sources. When you see κ in the videos and ξ in the papers, they refer to the same thing.

Further reading

1. Sections 3.1.4 and 3.1.5 of Coles (2001) for the max-stability derivation of the extremal types theorem and three closed-form examples (exponential gives Gumbel, Frechet gives Frechet, uniform gives Weibull). Section 3.3.3 for the return-level plot and how the shape parameter controls concavity.
2. Chapter 4 of Coles (2001) for the GPD and threshold models, including the four-line derivation of the GPD from the GEV (section 4.2.2). This is next week's primary source and pairs with Part IV of the HEC series.

Questions

Write a few sentences on each.

1. The Fisher-Tippett-Gnedenko theorem says that block maxima converge to one of three distributions depending on the parent population's tail. In your own words, what property of the parent distribution determines which of the three you get?

2. The HEC deck (slides 28–32) discusses reasons why annual maximum streamflow might not look GEV-distributed even though the theorem says it should. Name two of those reasons, and say which one you think is hardest to check for in practice.
3. Ghil et al. (2011) section 3.1 states the same theorem more concisely. What does the paper add, in its brief treatment, that the HEC deck does not cover?
4. The GEV shape parameter controls whether the distribution has a heavy tail, a light tail, or a bounded upper tail. For annual maximum streamflow at a site you know, which case do you think applies, and why?

Bibliography

Coles, S. (2001). *An Introduction to Statistical Modeling of Extreme Values*. London: Springer.

Ghil, M., Yiou, P., Hallegatte, S., Malamud, B. D., Naveau, P., Soloviev, A., et al. (2011). Extreme Events: Dynamics, Statistics and Prediction. *Nonlinear Processes in Geophysics*, *18*(3), 295–350. https://doi.org/10/fvzxvv